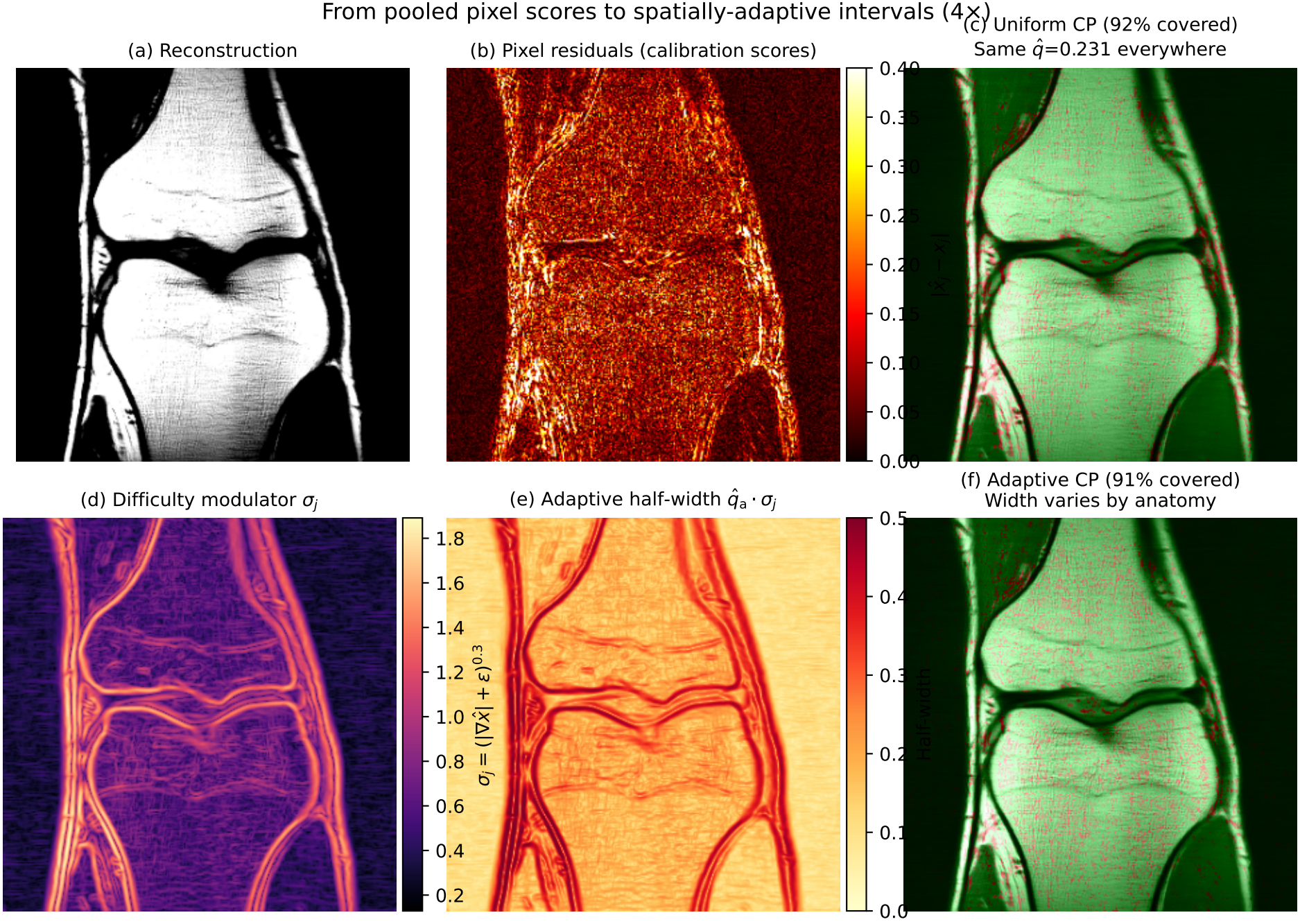
From pooled pixel scores to spatially-adaptive intervals (4×)
(a) Reconstruction
(b) Pixel residuals (calibration scores)
$|\hat{x}_j - x_j|$
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
(c) Uniform CP (92% covered)
Same $\hat{q}$=0.231 everywhere
(d) Difficulty modulator σ_j
$\sigma_j = (|\nabla \hat{x}| + \varepsilon)^{0.3}$
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
(e) Adaptive half-width $\hat{q}_a \cdot \sigma_j$
Half-width
0.5
0.4
0.3
0.2
0.1
0.0
(f) Adaptive CP (91% covered)
Width varies by anatomy